

Project Progress Report

Interpretable Sentiment Analysis of Customer Reviews Using Deep Learning

IE7615 Deep Learning for AI | Wiktoria Lasek

1. Summary of work completed so far

The project is progressing according to the proposed timeline. The following tasks have been completed:

- Data collection and loading:

Three datasets have been successfully loaded and explored. The Amazon Reviews [1] dataset (3600000 reviews) was loaded from a raw .txt file using a custom line-by-line parser. A stratified sample of 200000 training reviews and 40000 test reviews was selected to ensure computational efficiency while maintaining the original 50/50 class balance. The Yelp Reviews dataset [2] (38000 reviews, 5000 sampled) was loaded for cross-domain generalization testing.

- Data exploration:

Dataset	Rows	Columns	Missing values	Balance	Mean length
Amazon Train Set	3600000 (200k sample)	2 (label, review)	0	50/50	78 words
Amazon Test Set	400000 (40k sample)	2 (label, review)	0	50/50	78 words
Yelp (cross-domain)	38000 (5k sample)	2 (text, label)	0	50/50	133 words

- Data preprocessing:

Text preprocessing was applied for Logistic Regression and Feedforward Neural Network: lowercasing, punctuation removal, and stop-word removal using NLTK. Labels were converted from __label__1/ __label__2 format to binary 0/1. TF-IDF vectorization (max_features=10,000) was applied to convert clean text into numerical feature vectors.

- Model 1 - Logistic Regression (Baseline):

Logistic Regression was trained on 200000 Amazon reviews and evaluated on 40000 test reviews. LR coefficients were extracted for interpretability analysis -revealing the top 10 most positive and negative words learned by the model. The model was also tested on Yelp reviews for cross-domain evaluation.

2. Preliminary results

Model	Amazon accuracy	Amazon F1-Score	Yelp accuracy	Yelp F1-Score
Logistic Regression	88.82%	88.86%	87.86%	88.26%

The Logistic Regression baseline achieves a strong 88.82% accuracy – indicating that Amazon reviews contain clear, unambiguous sentiment language. The cross-domain test on Yelp shows only a 1% drop in accuracy (88.82% to 87.86%), suggesting the model learned genuine sentiment patterns rather than Amazon-specific vocabulary.

	Precision	Recall	F1-Score	Support
Negative (0)	0.92	0.83	0.87	2540
Positive (1)	0.84	0.93	0.88	2460
Overall	0.88	0.88	0.88	5000

Looking at the results by class, the model is better at correctly identifying negative reviews (precision 0.92) but catches more positive reviews overall (recall 0.93). This small difference is expected - Amazon product reviews and Yelp restaurant reviews are written differently, so the model is slightly less confident when predicting negative sentiment on Yelp data.

Most positive words	Most negative words
great (11.72), excellent (11.19), perfect (8.27), awesome (7.96), best (7.77), fantastic (7.64), amazing (7.42), love (7.26), highly (7.14), wonderful (7.07)	worst (-13.11), disappointing (-12.97), waste (-12.05), disappointment (-10.85), disappointed (-10.52), boring (-9.79), poor (-9.65), poorly (-9.38), terrible (-8.97), horrible (-8.73)

The LR coefficients provide a transparent, human-readable explanation of model decisions. Words like 'great', 'excellent' and 'perfect' strongly indicate positive sentiment while 'worst', 'disappointing' and 'waste' strongly indicate negative sentiment - exactly as expected from real customer reviews.

3. Updated timeline

Week	Task	Status
Week 1 (May 11-17)	Data collection and exploration	Completed
Week 2 (May 18-24)	Data preprocessing and TF-IDF	Completed
Week 3 (May 25-31)	Logistic Regression baseline	Completed
Week 4 (Jun 1-7)	FNN and LSTM [5]	In progress
Week 5 (Jun 8-11)	DistilBERT fine-tuning [6] + attention visualization	Upcoming
Week 6 (Jun 12-14)	Cross-domain testing + model comparison	Upcoming
Week 7 (Jun 15-22)	Final report and presentation	Upcoming

The project is on schedule. Weeks 1-3 have been completed as planned. LSTM and DistilBERT training will be conducted on the university HPC cluster to address GPU requirements.

4. Next Steps

- Week 4 - Feedforward Neural Network and LSTM: Implement and train the FNN using TF-IDF features with PyTorch. Then implement the bidirectional LSTM with GloVe 100d embeddings on the HPC cluster GPU. Compare results against the LR baseline.
- Week 5 - DistilBERT Fine-tuning: Fine-tune distilbert-base-uncased [4] on a stratified subset of 15,000 Amazon reviews using HuggingFace Transformers on the HPC cluster GPU. Generate attention weight heatmaps [3] for interpretability analysis and compare against LR coefficients.
- Week 6 - Cross-domain Testing and Model Comparison: Test the best performing model on the Yelp Reviews dataset. Generate final comparison table across all four models for both Amazon and Yelp datasets. Conduct ablation study.
- Week 7 - Final Report and Presentation: Write the final report, prepare presentation.

5. References

- [1] Amazon Customer Reviews Dataset. (2023). Kaggle.
<https://www.kaggle.com/datasets/bittlingmayer/amazonreviews>
- [2] Yelp Reviews Sentiment Dataset. (2022). Kaggle.
<https://www.kaggle.com/datasets/thedevastator/yelp-reviews-sentiment-dataset>
- [3] Vaswani, A., et al. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems, 30. <https://arxiv.org/abs/1706.03762>
- [4] Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of NAACLHLT 2019. <https://arxiv.org/abs/1810.04805>
- [5] Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735-1780.
<https://www.bioinf.jku.at/publications/older/2604.pdf>
- [6] Sanh, V., Debut, L. Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. <https://arxiv.org/abs/1910.01108>